

Viva questions

1. Which data structure did you use to represent the electric grid and why?

Graph using adjacency lists to model connections between grid nodes.

2. How is the graph implemented in your code?

Using `HashMap<String, GridNode>` with `List<GridNode>` for neighbors.

3. How does the code calculate the shortest or safest path in the grid?

By applying Dijkstra's algorithm based on failure probability.

4. What role does the PriorityQueue play in your implementation?

It selects nodes with the lowest failure probability for traversal.

5. Why is BFS used in the simulation, and where?

BFS helps explore node levels and detect cascading failures.

6. What are the time and space complexities of your shortest path algorithm?

Time: $O((V + E) \log V)$, Space: $O(V)$ using Dijkstra's algorithm.

7. How do you calculate the failure probability in each node?

Based on voltage, current, and temperature values.

8. How is the Queue data structure used in your project?

Used to manage event-driven updates and level-order processing.

9. How is the concept of dynamic data structures applied here?

Using ArrayList, HashMap, and PriorityQueue for flexibility.

10. What challenges did you face while applying DSA concepts?

Handling graph logic, queue behavior, and live data integration.

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1. What is a data structure?

A data structure is a way of organizing and storing data for efficient access and modification.

2. What are the types of data structures?

They are linear (like arrays, stacks) and non-linear (like trees, graphs).

3. What is the difference between an array and a linked list?

Array has fixed size and direct indexing; linked list has dynamic size and sequential access.

4. What is a stack and how does it work?

A stack is a LIFO (Last In First Out) data structure using push and pop operations.

5. What is a queue and where is it used?

A queue is a FIFO (First In First Out) structure used in scheduling and buffering.

6. What is a binary tree?

A binary tree is a hierarchical structure where each node has at most two children.

7. What is the difference between BFS and DFS?

BFS explores level by level using a queue; DFS explores depth-wise using a stack or recursion.

8. What is a hash table?

A hash table stores key-value pairs using a hash function for fast lookup.

9. What is the time complexity of binary search?

Binary search has a time complexity of $O(\log n)$ on a sorted array.

10. What is a graph in DSA?

A graph is a set of nodes (vertices) connected by edges, used to model networks.

11. What is recursion?

Recursion is a function calling itself to solve smaller instances of the problem.

12. What is a circular linked list?

A circular linked list connects the last node back to the first node.

13. What is the difference between linear and binary search?

Linear search checks each element; binary search divides the search space in half.

14. What is a doubly linked list?

A list where each node has both next and previous pointers.

15. What is the difference between stack and queue?

Stack is LIFO; queue is FIFO in element processing order.

16. What is a heap?

A heap is a complete binary tree used to implement priority queues.

17. What is the time complexity of inserting an element into a heap?

Insertion in a heap takes $O(\log n)$ time.

18. What is a trie?

A trie is a tree-based data structure used for efficient retrieval of strings.

19. What is dynamic programming?

Dynamic programming solves complex problems by breaking them into overlapping subproblems.

20. What is Big O notation?

Big O describes the worst-case time or space complexity of an algorithm.

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1. What was the objective of your project?

To predict electric grid failures using Java and DSA, and identify high-risk nodes for preventive action.

2. Which data structures did you use and why?

Graphs for grid modeling, Queues for event flow, and Priority Queues for risk-based processing.

3. How does the failure prediction system work?

It monitors voltage, current, temperature, and flags nodes with >30% failure probability.

4. What algorithms did you implement?

BFS and Dijkstra's algorithm to find safest paths to power generators.

5. How is weather data integrated into the system?

Weather impacts temperature at each node, affecting failure probability dynamically.

6. What tools and technologies did you use?

Java (JDK 8+), Visual Studio Code, and command-line for execution.

7. What challenges did you face during implementation?

Input validation, graph traversal logic, and weather-data integration were challenging.

8. Explain how graph traversal helps in failure prediction.

It helps find rerouting paths from high-risk nodes to generators using Dijkstra.

9. What are the real-world applications of your project?

Power companies can use it for real-time grid monitoring and failure prevention.

10. What did you learn from this project?

I learned Java implementation, graph algorithms, and practical use of simulation systems.